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CENTRAL DATA MODEL

& SYSTEM ARCHITECTURE

The shared foundation underneath all sixteen WHPS platform modules

Companion document to the WHPS Digital Enterprise Platform proposal and Product Specification

Build-Ready Data Model Draft | August 2026

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1. Purpose & Scope

The WHPS Digital Enterprise Platform proposal made the case for a connected system. The WHPS Product Specification broke that system into sixteen modules, each defined the same way across ten dimensions. This document takes the last step before code: it defines the twenty-two entities that sit underneath every module, how they relate to one another, which module owns each one, and the order in which they should be built.

This is the point where the platform stops being a set of module descriptions and becomes an actual schema — the reference an engineering team would use to design the database, define API contracts, and decide build sequence.

1.1 The Module Chain (recap)

Sixteen modules, one shared foundation underneath all of them:

#	Module
1	Public Website
2	Customer App / My Account
3	CRM
4	Sales / POS
5	Inventory
6	Branch Management
7	Franchise
8	HR + Recruitment
9	Finance
10	Procurement
11	Repairs
12	Custom Jewellery
13	Vendors / Karigars
14	Management Command Center
15	AI Intelligence
16	Central Admin

1.2 The One Architecture Rule That Matters Most

Every entity below has exactly one owning module — the only module allowed to create, update or delete it. Every other module that needs that data reads it through the owning module's API, never by writing to its tables directly.

This single rule is what turns sixteen modules into one connected system instead of sixteen disconnected applications. It is also what makes the Audit Log in Section 4 reliable: if only one module can ever write an entity, every change to that entity has exactly one place to be logged from.

2. Entity Ownership Matrix — Quick Reference

A single-page view of every entity, its owning module, and where else it is written or read. Full definitions, key fields and relationships for each entity follow in Section 3.

Entity	Owning Module	Also Writes
Users	Central Admin	HR + Recruitment
Customers	CRM	Public Website
Employees	HR + Recruitment	Branch Management
Branches	Central Admin	Branch Management
Franchises	Franchise	Finance
Products	Inventory	Public Website / App
Jewellery Items	Inventory	Sales
Inventory	Inventory	Sales
Sales	Sales	Finance
Orders	Public Website / Customer App (e-commerce layer)	Sales
Payments	Finance	Sales, Customer App
Leads	CRM	Public Website, Customer App, WhatsApp integration
Appointments	CRM	Customer App / Website
Repairs	Repairs	Inventory
Procurement	Procurement	Inventory
Vendors	Vendors / Karigars	Procurement
HR	HR + Recruitment	Central Admin
Finance	Finance	Mostly aggregates existing data; writes its own reconciliation and adjustment entries
Documents	Central Admin (storage and access control)	Sales
Notifications	Central Admin (shared notification service)	—
Audit Logs	Central Admin	—
AI Events	AI Intelligence Layer	—

“Also Writes” lists modules with a legitimate, specific write path into that entity (e.g. Sales updating an item’s status on purchase) — it is not general access. Every other module not listed here only reads the entity, and only within its own permission scope.

3. Entity Catalog

Every entity below follows the same structure: what it is, its core fields, how it relates to other entities, who owns it, who else writes to it, and who else can read it.

1. Users

Definition	Any authenticated actor in the system — internal staff, franchise partners and administrators. Distinct from Customers, who are the buying public and authenticate separately.
Key Fields	user_id, name, email, phone, role, branch_id / franchise_id (nullable), status, last_login, created_at
Relationships	• Belongs to one Branch or Franchise (optional) • Has one Role (permission set) • Has many Audit Log entries and Notifications • Linked 1:1 to one Employee record where applicable
Owning Module	Central Admin
Also Writes	HR + Recruitment (creates the linked User account automatically when a candidate is hired)
Read-Only Access	Every module (for authentication and permission checks)

2. Customers

Definition	Any individual who has purchased, enquired with, or registered on WHPS, tracked as one unified profile across every channel.
Key Fields	customer_id, name, phone, email, addresses, DOB / anniversary, preferences, consent flags, lifetime value, assigned salesperson, loyalty tier
Relationships	• Has many Leads, Appointments, Orders / Sales, Repairs, Payments, Documents (invoices, certificates) • Linked to one primary Branch (home store)
Owning Module	CRM
Also Writes	• Public Website (signup) • Customer App (profile updates) • Sales (purchase history updates on each transaction)
Read-Only Access	• Franchise (where permitted by agreement) • Management Command Center, AI Intelligence

3. Employees

Definition	The WHPS staff record, created automatically once a candidate is hired, and referenced across HR, branch and sales modules.
Key Fields	employee_id, name, contact details, role / designation, branch_id, joining_date, reporting_manager, employment_status
Relationships	• Linked 1:1 to one User account • Belongs to one Branch or Franchise • Has many Attendance / Leave / Performance records • Has many Sales records (where the employee is a salesperson) • Originates from one Candidate (HR) record
Owning Module	HR + Recruitment
Also Writes	• Branch Management (role / shift assignment) • Central Admin (account status changes)
Read-Only Access	Sales, CRM, Branch Management, Management Command Center

4. Branches

Definition	A WHPS-operated showroom location.
Key Fields	branch_id, name, address, region, manager_id, operating_hours, status
Relationships	• Has many Employees, Inventory records, Sales, Appointments, Leads • Belongs to one Region
Owning Module	Central Admin
Also Writes	Branch Management (day-to-day operational updates)
Read-Only Access	Nearly every module — branch_id is a near-universal foreign key across the platform

5. Franchises

Definition	An independently owned WHPS-branded showroom operating under a franchise agreement.
Key Fields	franchise_id, owner / applicant details, location, agreement_start / end, territory, commercial terms, status
Relationships	• Originates from one Franchise Application • Has many Employees, Inventory allocations, Sales • Belongs to one Region

Owning Module	Franchise
Also Writes	• Finance (royalty / fee records) • Inventory (allocation records)
Read-Only Access	Management Command Center, AI Intelligence

6. Products

Definition	The catalogue-level definition of a jewellery design (a SKU), independent of any single physical piece.
Key Fields	product_id, category, collection, design, metal, purity, making-charge rule, base images, price rule
Relationships	• Has many Jewellery Items (the physical instances of this design) • Belongs to one Category / Collection
Owning Module	Inventory
Also Writes	• Public Website / App (display content) • Custom Jewellery (creates a new product definition from an approved custom design)
Read-Only Access	Sales, CRM, Franchise, Management Command Center

7. Jewellery Items

Definition	One individual physical piece — the Digital Passport instance of a Product, tracked from receipt through sale and any after-sales event.
Key Fields	item_id, product_id, HUID, gross / net weight, stone details, cost, status, current_branch or franchise_id, vault / showcase location
Relationships	• Belongs to one Product • Belongs to one Branch or Franchise (current location) • Has many Movement / Transfer records • Linked to one Sale (when sold) • Linked to Repair records (if serviced)
Owning Module	Inventory
Also Writes	• Sales (status change on sale) • Repairs (status change) • Franchise (allocation) • Custom Jewellery (new item creation on completion)

Read-Only Access	Sales, CRM, Franchise, Management Command Center, AI Intelligence
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8. Inventory

Definition	The status-and-location ledger layered on top of Jewellery Items — what is available, reserved or in transit at any moment.
Key Fields	inventory_id, item_id, branch_id or franchise_id, status, last_reconciled_at
Relationships	• Belongs to one Jewellery Item • Belongs to one Branch or Franchise • Has many Transfer records • Has many Reconciliation records
Owning Module	Inventory
Also Writes	• Sales (reservation / sale) • Procurement (goods receipt) • Repairs (status)
Read-Only Access	Sales, Franchise, Management Command Center, AI Intelligence

9. Sales

Definition	A transaction record — quotation through final invoice — for a specific customer and one or more items.
Key Fields	sale_id, customer_id, branch_id or franchise_id, salesperson_id, item_ids, amount, discount, payment_status, invoice_number
Relationships	• Belongs to one Customer • Belongs to one Branch or Franchise • References one or more Jewellery Items • Has one or more Payments • May originate from one Lead / Quotation
Owning Module	Sales
Also Writes	• Finance (revenue recognition) • CRM (loyalty points) • Inventory (status update)
Read-Only Access	CRM, Finance, Management Command Center, AI Intelligence

10. Orders

Definition	An online or app-originated purchase or reservation — distinct from a Sale, which is the underlying transaction record it eventually converts into.
Key Fields	order_id, customer_id, sale_id (once converted), fulfilment_type, delivery_address, status, created_channel
Relationships	• Belongs to one Customer • Converts into one Sale on completion • Has one Payment • Has one Fulfilment / Delivery record
Owning Module	Public Website / Customer App (e-commerce layer)
Also Writes	• Sales (order-to-sale conversion) • Inventory (stock allocation on order placement)
Read-Only Access	CRM, Branch Management, Management Command Center

11. Payments

Definition	Any monetary transaction recorded against a Sale, Order, gold-scheme instalment, or Repair.
Key Fields	payment_id, related_entity_type, related_entity_id, amount, method, status, timestamp, gateway_reference
Relationships	• Belongs to one Sale, Order, Scheme instalment, or Repair • Belongs to one Customer
Owning Module	Finance
Also Writes	Sales, Customer App (scheme payments), Repairs
Read-Only Access	CRM, Management Command Center

12. Leads

Definition	Any expression of customer interest not yet converted to a sale — a website enquiry, a WhatsApp message, or a walk-in.
Key Fields	lead_id, customer_id (once matched), source, status, assigned_salesperson, budget / interest notes
Relationships	• Belongs to one Customer (once matched or created) • Belongs to one Branch • May convert into one Appointment or one Sale

Owning Module	CRM
Also Writes	Public Website, Customer App, WhatsApp integration (lead capture)
Read-Only Access	Sales, Management Command Center, AI Intelligence

13. Appointments

Definition	A scheduled store visit for a defined purpose — bridal shopping, repair drop-off, consultation, and similar.
Key Fields	appointment_id, customer_id, branch_id, date_time, purpose, status
Relationships	• Belongs to one Customer • Belongs to one Branch • May originate from one Lead • May result in one Sale
Owning Module	CRM
Also Writes	• Customer App / Website (self-service booking) • Branch Management (scheduling)
Read-Only Access	Sales, Management Command Center

14. Repairs

Definition	A service ticket for repair, resizing, or maintenance work on a customer-owned or WHPS-owned item.
Key Fields	repair_id, customer_id, item_reference, branch_id, karigar_id, status, estimate, sla_due_date
Relationships	• Belongs to one Customer • Belongs to one Branch • Assigned to one Karigar / Vendor • Has one or more Payments
Owning Module	Repairs
Also Writes	• Inventory (item status = under repair) • Customer App (status updates)
Read-Only Access	CRM, Management Command Center

15. Procurement

Definition	The purchase-requisition-to-goods-receipt record used to restock inventory from vendors.
Key Fields	po_id, vendor_id, requested_by, items_and_quantities, status, expected_delivery, qc_result
Relationships	• Belongs to one Vendor • Belongs to one Branch or Corporate entity • Results in Inventory goods-receipt records
Owning Module	Procurement
Also Writes	• Inventory (stock on receipt) • Finance (payables)
Read-Only Access	Management Command Center, AI Intelligence

16. Vendors

Definition	An external supplier of finished goods, raw material, or manufacturing / karigar services.
Key Fields	vendor_id, name, type (supplier / karigar), contact, performance_rating, contract_terms
Relationships	• Has many Procurement orders • Has many Karigar job-cards (where applicable) • Has many Payments (payables)
Owning Module	Vendors / Karigars
Also Writes	• Procurement • Custom Jewellery (job assignment)
Read-Only Access	Finance, Management Command Center

17. HR

Definition	The recruitment-to-employment lifecycle data — candidates, applications, assessments, interviews — that precedes and then feeds the Employee record.
Key Fields	candidate_id, application_data, assessment_scores, interview_scorecards, offer_status
Relationships	• Converts into one Employee record on joining • Belongs to one Job Opening • Has many Interview records
Owning Module	HR + Recruitment

Also Writes	Central Admin (creates the new Employee's User account)
Read-Only Access	Management Command Center

18. Finance

Definition	The consolidated financial ledger view — revenue, refunds, discounts, payables / receivables, royalty — drawing from Sales, Payments, Procurement and Franchise.
Key Fields	ledger_entry_id, type, related_entity, amount, branch_id or franchise_id, date, reconciliation_status
Relationships	Aggregates Sales, Payments, Procurement and Franchise royalty records
Owning Module	Finance
Also Writes	Mostly aggregates existing data; writes its own reconciliation and adjustment entries
Read-Only Access	Management Command Center, AI Intelligence

19. Documents

Definition	Any stored file or certificate tied to a customer, employee, franchise, or item — invoices, HUID certificates, KYC documents, agreements.
Key Fields	document_id, owner_entity_type, owner_entity_id, type, file_reference, verification_status, uploaded_by
Relationships	Belongs to one Customer, Employee, Franchise, or Jewellery Item
Owning Module	Central Admin (storage and access control)
Also Writes	• Sales (invoices) • Inventory (certificates) • HR (candidate documents) • Franchise (agreements)
Read-Only Access	Every module holding a reference to a document

20. Notifications

Definition	Any outbound message triggered by a system event — email, SMS, WhatsApp, or push.
Key Fields	notification_id, recipient, channel, event_type, related_entity, status, sent_at
Relationships	• Triggered by an event in any module • Belongs to one Customer, Employee, or Applicant

Owning Module	Central Admin (shared notification service)
Also Writes	— (system-generated on behalf of every module)
Read-Only Access	Management Command Center (delivery reporting)

21. Audit Logs

Definition	An immutable record of who changed what, when, and from what value to what value, across every sensitive action on the platform.
Key Fields	log_id, user_id, action, entity_type, entity_id, before_value, after_value, timestamp, approved_by
Relationships	• References any entity in the system • Belongs to one User (the acting party)
Owning Module	Central Admin
Also Writes	— (written automatically by every module on sensitive actions)
Read-Only Access	Management Command Center, Central Admin audit viewer

22. AI Events

Definition	A logged AI recommendation or prediction, together with its confidence, evidence and the human decision made on it — the record that keeps AI accountable.
Key Fields	event_id, model_or_feature, related_entity, recommendation, confidence, evidence_summary, human_decision, decided_by
Relationships	• References any entity (an Inventory transfer, a Lead, a Franchise application, etc.) • Belongs to one reviewing User
Owning Module	AI Intelligence Layer
Also Writes	— (written by the AI layer; decisions recorded by the relevant module owner)
Read-Only Access	Management Command Center, Central Admin

4. Core Relationship Map

Rather than one dense diagram, the twenty-two entities group naturally into five clusters. Understanding these clusters matters more than any single foreign key — it tells you which entities change together and which module boundaries are safe to build independently.

4.1 People & Organisation Cluster

Users → Employees → Branches / Franchises, with HR feeding Employees and Central Admin owning Users. This cluster answers “who is this person and where do they work” for every other entity in the system.

4.2 Customer Cluster

Customers sit at the centre, with Leads, Appointments, Orders, Sales, Repairs, Payments and Documents all referencing back to a single Customer record. This is the cluster the Customer 360° profile is built from.

4.3 Product & Inventory Cluster

Products → Jewellery Items → Inventory, with every physical item traceable back to its catalogue definition and forward through every location and status change it has ever had. Procurement and Vendors feed this cluster from the supply side.

4.4 Transaction Cluster

Sales, Orders, and Payments reference Customers and Jewellery Items, and in turn feed Finance. This is the cluster with the tightest consistency requirements — a Sale, its Payment and the Inventory status change it triggers should always move together.

4.5 Platform Cluster

Documents, Notifications, Audit Logs and AI Events are cross-cutting — every other cluster writes into them, but they don't belong to any single business process. These four are infrastructure entities, owned by Central Admin and the AI Intelligence Layer, and should be built early even though they have no screens of their own.

5. Suggested Build Order by Data Dependency

This is a dependency order, not a delivery timeline — it says what has to exist before something else can be built correctly, not how long each step should take. Duration, staffing and sequencing against business priorities belong in a separate delivery plan agreed with WHPS.

Tier 1 — Foundation

Nothing else can be built correctly without these. They have no dependencies on other entities in this list.

- Users, Branches (Central Admin)
- Documents, Notifications, Audit Logs (Central Admin — infrastructure, no screens of their own)

Tier 2 — Master Data

Depends only on Tier 1. These are the “noun” entities every later transaction will reference.

- Customers (CRM)
- Employees (HR + Recruitment, depends on Users)
- Products (Inventory)
- Vendors (Vendors / Karigars)
- Franchises (Franchise, depends on Branches for territory logic)

Tier 3 — Physical & Relationship Layer

Depends on Tier 2. This is where individual jewellery items and customer relationships become trackable.

- Jewellery Items and Inventory (Inventory, depends on Products + Branches/Franchises)
- Leads and Appointments (CRM, depends on Customers + Branches)
- HR candidate pipeline (HR + Recruitment, depends on Employees' target structure)

Tier 4 — Transactions

Depends on Tier 3 being live and reliable. This is the commerce core of the platform.

- Sales (Sales, depends on Customers + Jewellery Items + Employees)
- Orders (Public Website / Customer App, depends on Customers + Sales)
- Payments (Finance, depends on Sales / Orders)
- Procurement (Procurement, depends on Vendors + Inventory)
- Repairs (Repairs, depends on Customers + Jewellery Items)

Tier 5 — Oversight & Intelligence

Depends on Tier 4 producing real data to aggregate, forecast and score against.

- Finance ledger aggregation (Finance)
- Management Command Center (reads across every module)
- AI Events and the AI Intelligence Layer (needs transaction history before any model is useful)

A module can start UI and API scaffolding ahead of its tier, but should not go live with real data until the entities it depends on are stable — otherwise foreign keys get built against data models that are still changing.

6. System Architecture Principles

6.1 One Owning Module per Entity

Restated from Section 1.2 because it is the single most important technical decision in this document: each entity has exactly one module with write authority. Every cross-module interaction is an API call to the owning module, never a direct write to its data.

6.2 Central Data Platform as a Shared Layer, Not a Separate Product

The Central Data Platform referenced throughout the proposal and specification is not a 17th module — it is the shared database and API layer that Sections 2–5 define. Whether it is implemented as one database with modular services or as separate services each owning their own tables, the entity boundaries and ownership rules in this document should hold either way.

6.3 Every Sensitive Write Produces an Audit Log Entry

Any create, update or delete on Customers, Jewellery Items, Sales, Payments, Franchise agreements, or Employee records should write a corresponding Audit Log entry in the same transaction — not as an afterthought process that can silently fail.

6.4 AI Reads Everywhere, Writes Almost Nowhere

The AI Intelligence Layer should have broad read access across the Central Data Platform to generate recommendations, but its only direct write target is its own AI Events log. Any action an AI recommendation leads to — a transfer, a discount, a franchise approval — is written by the owning module, only after a human approves it there.

6.5 Notifications and Documents Are Infrastructure, Not Features

Treat Notifications and Documents as shared services every module calls, not as something each module reimplements. A Repair-ready SMS and a Franchise-agreement-renewal alert should go through the same notification service; a HUID certificate and a franchise agreement PDF should go through the same document store, differing only in `owner_entity_type`.

7. Next Steps

1. Turn Section 3's "Key Fields" lists into an actual schema — data types, nullability, indexes — for each of the twenty-two entities.
2. Define the API contract for each owning module: what it exposes for other modules to read, and what write endpoints it allows for the "Also Writes" relationships in Section 2.
3. Confirm the physical inventory model with WHPS — whether every Jewellery Item is fully serialised (one row per physical piece) or whether some categories need quantity-based stock, since this affects the Inventory schema directly.
4. Validate the Tier 1–5 build order in Section 5 against WHPS's actual priorities (e.g. if Franchise is not an immediate priority, Tier 2–4 can proceed without it).
5. Decide the underlying technical architecture — modular monolith vs. independent services — using the ownership boundaries in Sections 2 and 3 as the service/module split either way.
6. Design the Audit Log and Notification service first among the infrastructure entities, since every later module will depend on calling them correctly from day one.

With this data model in place alongside the platform proposal and the product specification, WHPS has what is needed to move from requirements into schema design, API definition, and the first lines of code.